

Stress Testing Ed-HMI220-070C

Nyle Controls Group

March 27, 2026

Abstract

In an effort to better detect behaviours of products at the limits of our operational range, we are stress testing devices at max loads for extended periods of time. **TLDR: HMI Max temp:87.1 °C (188.78 °F) while in a sealed and insulated box** The screen may behave like a heat sink, and orientation may impact cooling. at 100% load the HMI trends towards 61°C (140 °F)

Method

The Ed-HMI220-070C in this testing sequence was subjugated to 100% CPU load for the duration of the workday 3.23.26 packed in an insulated box with 0 ventilation.

At the end of the day the Ed-HMI220-070C was removed from that box, and left running at 100% CPU load overnight.

Tools

```
$ sudo apt install stress-ng lm-sensors
```

also get:

```
https://github.com/elovejoy-nyle/loadTest/blob/main/cpu\_burn.sh
```

which is a modified script from the authors of stress-ng package, with inclusions of logging.

lm-sensors was pre-configured on the distro which shipped with this device, if it were not, you would run:

```
sudo sensors-detect --auto
```

Results

HMI Max temp:87.1 °C (188.78 °F) while in an insulated, unventilated box.

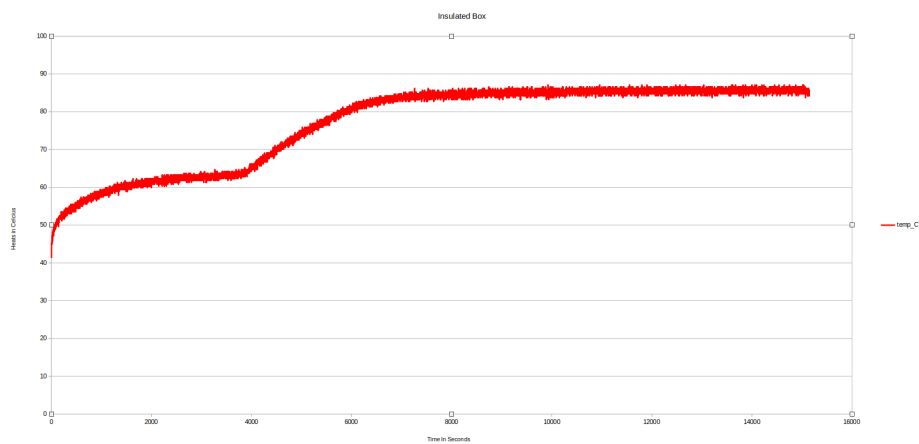


Figure 1: HMI in insulated box. Bump is where lid was closed

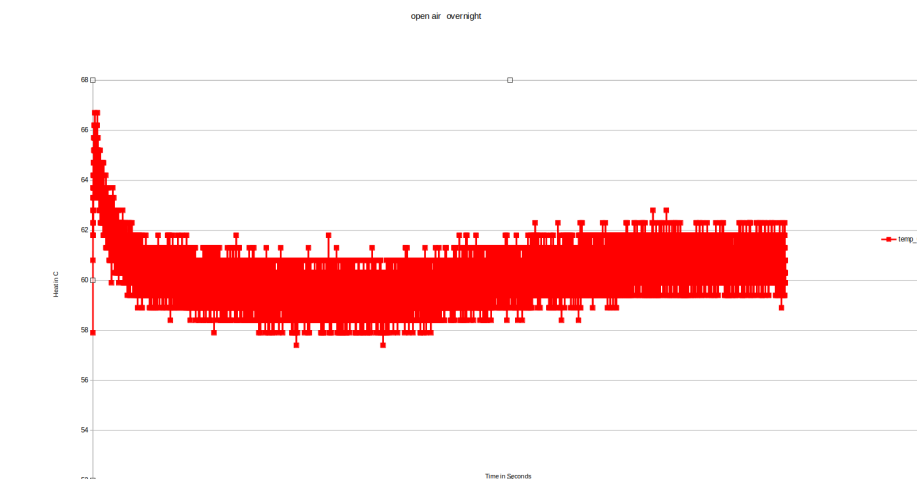


Figure 2: Test began immediately after insulated box testing

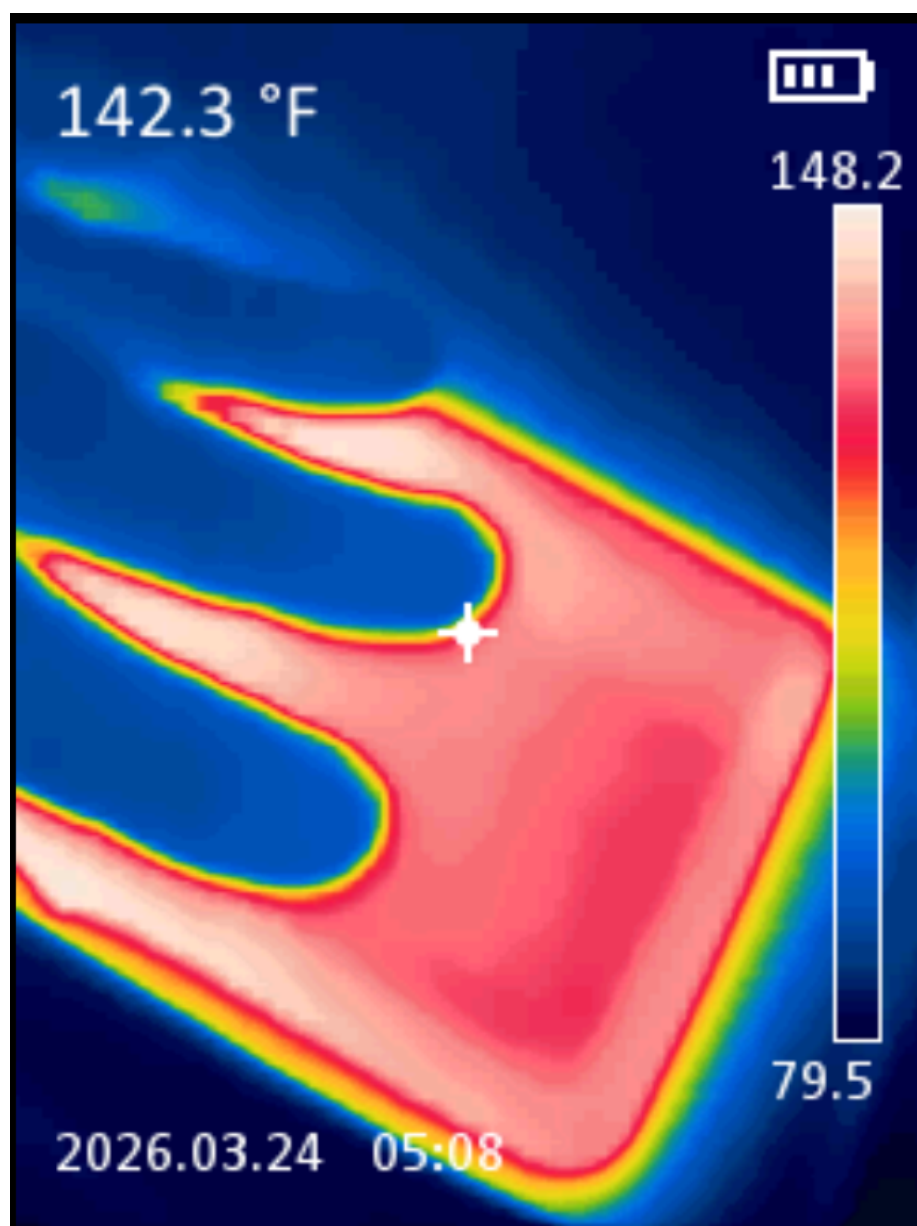


Figure 3: Human hand for contrast, after insulated test